

Effect of variety, processing, and storage on the flavonoid glycoside content and composition of lettuce and endive.



Eight varieties of lettuce (*Lactuca sativum*) and three varieties of endive (*Cichorium endivia*) were analyzed for flavonoid composition and content. Total flavonoid contents, expressed as units of aglycon for fresh material, were in the ranges of 0.3-229 microg/g for lettuce and 44-248 microg/g for endive. Five quercetin conjugates [quercetin 3-O-galactoside, quercetin 3-O-glucoside, quercetin 3-O-glucuronide, quercetin 3-O-(6-O-malonyl)glucoside, and quercetin 3-O-rhamnoside] and luteolin 7-O-glucuronide were measured in the green-leafed lettuce and an additional two cyanidin conjugates [cyanidin 3-O-glucoside and cyanidin 3-O-[(6-O-malonyl)glucoside]] in the red-leafed varieties. Three kaempferol conjugates [kaempferol 3-O-glucoside, kaempferol 3-O-glucuronide, and kaempferol 3-O-[6-O-malonyl)glucoside]] were measured in each of the endive varieties. The presence and identity of kaempferol 3-O-(6-O-malonyl)glucoside in endive was shown for the first time. Shredding of lettuce leaf followed by exposure to light produced significant losses of the flavonoid moiety in the green oak leaf (94%), red oak leaf (43%), iceberg (36%), green batavia (25%), lollo biondo (24%), and lollo rosso (6%) samples, whereas cos and green salad bowl samples did not show an overall loss. Shredding of endive also produced loss of the flavonoid moiety in escarole (32%), fine frisee (13%), and coarse frisee (8%). Significant demalonation was observed for both the quercetin and cyanidin glucosides in lettuce, whereas a similar degradation of the kaempferol analogue was found in endive tissue. Storage of whole heads of both lettuce and endive in the dark at 1 degrees C and 98% humidity for 7 days resulted in losses of total flavonol glycosides in the range of 7-46%. The identification of the amounts, position of substitution, and nature of the sugars is important for understanding the potential bioavailability and biological activities of flavonoids in salads.